

# eSpace for many schools

How eSpace becomes one system that many schools use - each with its own space, all sharing the content that St Mark's built.

Prepared 5 October 2026 · For the eSpace team · Status: agreed direction, to be built after the current round of improvements

## 1. In short

- **One system, many schools.** Every school uses the same eSpace, at its own web address (for example `stmark.<espace-domain>`, `kings.<espace-domain>`).
- **Each school is separate.** Every school has its own database: its students, teachers, marks, chats and reports are never visible to another school.
- **St Mark's is the master school.** Its teaching content - curriculum, eNotes, eLibrary, Item Bank, Virtual Lab - is the master copy. It also stays our test school: new features go to St Mark's first.
- **New schools start with St Mark's content.** Their teachers can read it and edit it. Edits stay in their school only; the master at St Mark's never changes.
- **Master updates flow out.** When St Mark's improves a topic, schools that have not edited it get the new version automatically; schools that edited it are told and can choose.
- **A developer console** - only for the eSpace team - to enrol, import, suspend and delete schools, and update them all at once.

## 2. Decisions already made

| Question                              | Decision                                                                                                                                                                                                                                                                                                             |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| How are schools kept apart?           | <b>One database per school</b> , all running the same code. The school is chosen from the web address. Rejected: one shared database with a school column on every table (too risky - one missed filter shows another school's students); a separate copy of the code per school (every update deployed many times). |
| Where does shared content come from?  | <b>St Mark's</b> is the master school.                                                                                                                                                                                                                                                                               |
| Can other schools change the content? | <b>Yes - in their own school only.</b> The first edit makes the item "edited locally"; the master is untouched.                                                                                                                                                                                                      |
| How do schools get the content?       | <b>Copied into each school</b> when it is enrolled, and kept in step by a sync (section 5). Not read live from St Mark's: if St Mark's deletes something, other schools keep theirs, and each school stays fast and independent.                                                                                     |

|                       |                                                                                                                                              |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| What is never shared? | Anything about people: students, teachers' accounts, marks and submissions, Learning Maps, chats, notices, parent links, report cards, logs. |
| Domain                | eSpace gets its own new domain (to be chosen). Schools get a subdomain of it.                                                                |

### 3. To settle before we build

1. **Content agreement with St Mark's.** Most master content was written by St Mark's teachers, and St Mark's is a client. Agree in writing that content made on eSpace may be shared with other eSpace schools - for example with credit ("Notes by St Mark's College", shown automatically) and/or a discount for St Mark's.
2. **The domain name** - buy it, and set a wildcard subdomain ( `*.<domain>` ) pointing to the eSpace folder.
3. **Hosting** - check how many databases the hosting plan allows, and create a cPanel API token so the console can create a school's database by itself. (Without a token: create each database by hand in cPanel and paste its name into the console.)
4. **Pricing** - price per learner per term (already on the website), trial length, and what happens when a school stops paying (suspend after N days).
5. **Email for the platform** - an address for school-admin invitations and the weekly parent digest across all schools.

### 4. How it works - architecture

#### 4.1 Finding the school

- Every request looks at the web address: `kings.<domain>` → school "kings" → its database.
- A small **platform database** holds the list of schools and how to reach each one's database. It holds no student data.
- The main website ( `<domain>` , no subdomain) shows the public landing site and the guide, as now.
- Locally we test with `stmark.localhost` and `demo.localhost` - no domain needed to build it.

#### 4.2 What changes in the code

| Area                        | Change                                                                                                                                                                                    |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Database connection         | Chosen per request from the school's record, instead of one fixed database in <code>.env</code> .                                                                                         |
| Uploaded files              | Each school's uploads in its own folder: <code>uploads/&lt;school&gt;/...</code> . Shared master files (book PDFs, images in notes) referenced, not copied, where possible to save space. |
| Sign-in                     | Sessions tied to the school's subdomain, so a login at one school never works at another.                                                                                                 |
| School name, crest, colours | Already per school ( <code>school_settings</code> ) - just lives in each school's database.                                                                                               |

|                 |                                                                                                                    |
|-----------------|--------------------------------------------------------------------------------------------------------------------|
| Scheduled jobs  | The weekly parent digest (and future jobs) runs once for every active school.                                      |
| Everything else | Unchanged: every page keeps working as it does today, because each school's database looks exactly like St Mark's. |

### 4.3 Platform database (new)

| Table             | Holds                                                                                                                                                              |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| platform_admins   | The eSpace team's console logins (separate from any school admin).                                                                                                 |
| schools           | Name, subdomain, database name, status (trial / active / suspended), plan, learner limit, contact, created and last-active dates, whether it is the master school. |
| school_migrations | Which database updates each school has had - so "update all schools" knows what to run.                                                                            |
| content_sync_log  | Each push of master content: when, what, which schools took it.                                                                                                    |
| platform_audit    | Who in the team enrolled, suspended or deleted which school, and when.                                                                                             |

## 5. Shared content - copy and keep in step

### 5.1 What is shared

#### Shared from St Mark's (content)

- Curriculum: topics, learning outcomes, competencies, constructs
- eNotes: topics, pages, narration, images
- eLibrary books and Item Bank papers
- Videos (links / files)
- Virtual Lab experiments and templates
- Question bank and AOI scenarios (optional)

#### Never shared (people and results)

- Students, teachers, HODs, admins
- Submissions, marks, Learning Maps, report cards
- Chats, notices, parent links, notifications
- Reading progress, notes, highlights
- Logs and settings

### 5.2 How each copied item is tracked

Every shared content row in a school's database gets four small fields:

| Field          | Meaning                                                                 |
|----------------|-------------------------------------------------------------------------|
| master_id      | The item's id at St Mark's (empty for the school's own new items).      |
| master_version | Which version of the master this copy is based on.                      |
| edited_locally | Set the first time a teacher at this school changes the item.           |
| source_credit  | "St Mark's College Namagoma" - shown on the item if agreed (section 3). |

### 5.3 What happens when ...

| Event                                    | Result                                                                                                                      |
|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| A school is enrolled                     | It receives a copy of the master content packs we choose (all, or by subject / class level).                                |
| A teacher at another school edits a note | Their copy becomes "edited locally". St Mark's master is untouched. Other schools are untouched.                            |
| A school adds its own new note           | It is theirs only (no <code>master_id</code> ).                                                                             |
| St Mark's improves a master item         | Its version goes up. At the next sync, schools that have <b>not</b> edited it get the new version automatically.            |
| ... and a school <b>had</b> edited it    | They keep their version and see a notice: "St Mark's updated this topic - compare / switch to the new version / keep mine". |
| St Mark's deletes a master item          | Other schools keep their copy (it simply stops receiving updates).                                                          |
| Sync runs                                | From the console ("Push master updates"), or on a schedule. Each run is logged.                                             |

## 6. The developer console

A separate area, only for the eSpace team, at `console.<domain>`.

| Screen             | What it does                                                                                                                                                                                                        |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Schools            | Every school: status, learners, teachers, storage used, last activity, plan. Search and filter.                                                                                                                     |
| Enrol a school     | Name, subdomain, crest, colours, first admin's name and email, plan. Creates the database from the template, runs all migrations, copies the chosen content packs, creates the admin account, sends the invitation. |
| School page        | Details, usage, the content it has edited, its migration status; actions: open as support (logged), suspend / reactivate, delete.                                                                                   |
| Suspend            | Blocks sign-in with a friendly message; keeps all data.                                                                                                                                                             |
| Delete             | Takes and downloads a full backup first, asks for the school's name to be typed, then removes the database and files. Logged.                                                                                       |
| Update all schools | Runs new database migrations on every school, one by one, with a dry run first and a result per school.                                                                                                             |
| Master content     | What St Mark's has changed since the last push; push to all schools or chosen ones.                                                                                                                                 |
| Usage and billing  | Learners per school per term, for invoicing; trial end dates; overdue schools.                                                                                                                                      |
| Team               | Console accounts for the four of us, and the audit log.                                                                                                                                                             |

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## 7. Onboarding a new school

1. **Enrol** in the console (section 6) - the school's space exists within minutes.
2. **Set up the year:** academic year, terms, departments, subjects (pre-filled from St Mark's, editable).
3. **Classes and streams in bulk:** e.g. S.1-S.6 × East / West / North in one step.
4. **Import staff** from Excel/CSV: name, email/phone, department, role (teacher / HOD), subjects. Preview with errors highlighted before saving; logins created.
5. **Import students** from Excel/CSV: name, admission number, class, stream, gender. Preview, then save; usernames and first passwords created and printable as login slips.
6. **Assign teachers** to classes and subjects.
7. **Check and hand over:** a checklist page confirms each step, then the school admin takes over. Training follows module by module.

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## 8. Build plan

| # | Phase               | Delivers                                                                                                                                                                               | Estimate       |
|---|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| 1 | School-aware eSpace | School chosen by web address; per-school database connection, uploads folder and sessions; platform database; St Mark's set up as school #1 (master). Tested locally with two schools. | 2-3 days       |
| 2 | Developer console   | Console login; schools list; enrol (database from template, migrations, admin account); suspend; delete with backup; update all schools; audit log.                                    | 3-4 days       |
| 3 | Onboarding tools    | Bulk classes and streams; staff and student import with preview and login slips; set-up checklist.                                                                                     | 2-3 days       |
| 4 | Shared content      | Master tracking fields; copy content packs at enrolment; "edited locally"; push master updates; "St Mark's updated this" notice with compare / switch / keep; credit line.             | 3-4 days       |
| 5 | Usage and billing   | Learners per school per term; trial and overdue view; suspend-after-overdue rule.                                                                                                      | 1-2 days       |
| 6 | Go-live             | New domain and wildcard subdomain; St Mark's moved to <code>stmark.&lt;domain&gt;</code> (old address redirects); first new school enrolled.                                           | 1 day + checks |

Total: about 2-3 weeks of work, built and tested phase by phase on localhost. Nothing changes for St Mark's until phase 6.

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## 9. Safeguards

- **Isolation tests:** automated checks that a login, link or file from one school never opens another school's data.
- **Backups:** every school backed up on a schedule; always before a delete or a migration run.
- **Migrations:** dry run first; run school by school; stop and report on the first failure; never leave a school half-updated.

- **Support access:** the team can open a school only through the console, and every visit is logged.
  - **Content sync:** never overwrites a school's local edits; every push logged and reversible for unedited items.
  - **Performance:** each school is its own database, so one busy school does not slow the others; watch total server load as schools grow.
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## 10. Go-live checklist

- ☐ Content-sharing agreement signed with St Mark's
  - ☐ Domain bought; wildcard subdomain pointing to eSpace
  - ☐ cPanel API token created (or manual database steps agreed); database limit checked
  - ☐ Platform email address set up
  - ☐ Phases 1-5 tested locally with St Mark's plus two demo schools
  - ☐ Full backup of St Mark's taken
  - ☐ St Mark's moved to its new address; old address redirects; St Mark's admin informed
  - ☐ First new school enrolled, imported and handed over
  - ☐ Pricing, trial and suspension rules written down
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eSpace · Multi-school plan · Keep this document in `docs/` and update it when a decision changes.